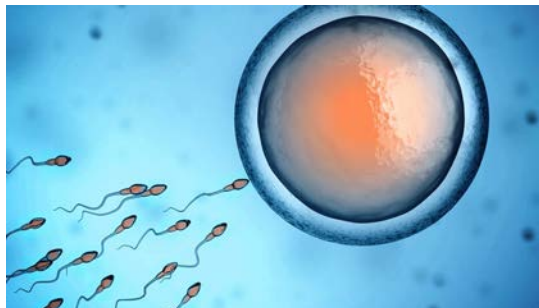
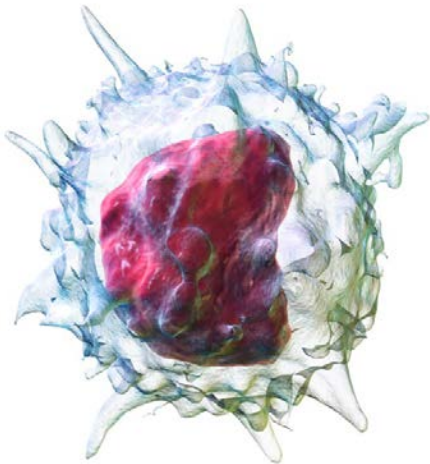
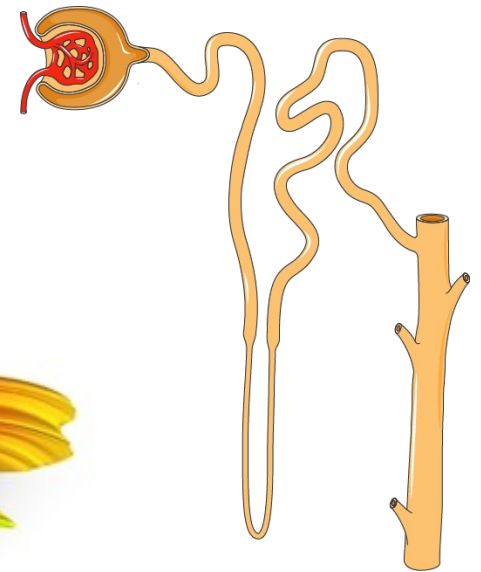
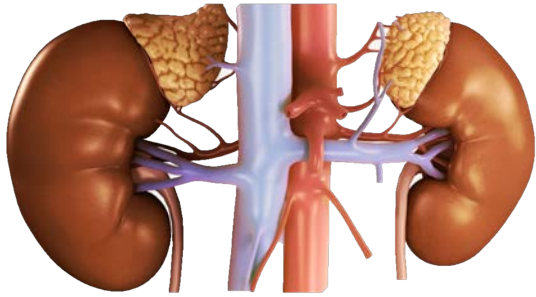
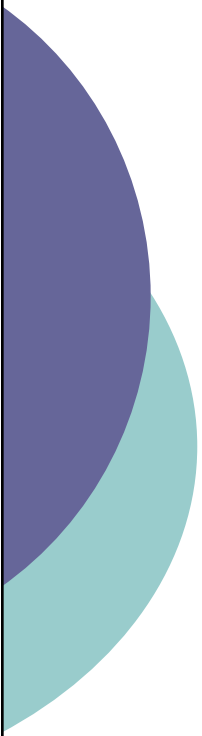


Welcome to Bio 40C!

Jeff Schinske



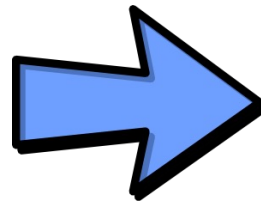


Bio 40C Proposed Norms

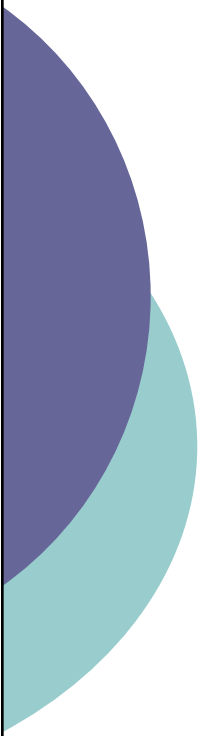
- Seek to enjoy your colleagues and your experiences in class!
- Actively select language that is inclusive, respectful, and welcoming
- Assume good intentions...no heroes and no villains (in this class)
- Recognize that we all have strengths as well as areas for growth
- Be curious, be skeptical
- Ask for help (and expect to receive it!)

Share with a neighbor...

1. Name
2. Something important to know about who you are
3. What most stood out to you from Week 1 of Bio 40C



Person with 1st name closest to end of alphabet goes first



Debriefing Last Week

- Your Responses: End-of-Class Index Card
 - Some examples!
- Last Week's Canvas Work
 - "This weeks online work was a little challenging for me. I think it is due to being the first week so I'm just getting back into the groove."
 - "At first, it seemed like a lot of online work in the week 1 module by a glance. However, when I started to do it, I just couldn't stop. The structure of the module is coherent and reasonable."
 - "After learning 40a and 40B , I am able to understand so much better and make connections."
 - "I enjoyed listening to Rob Lim's heroic story and its outcome!"



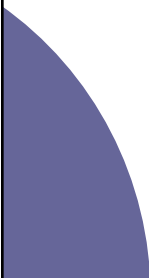
Some of Your Thoughts: Robert Lim Scientist Spotlight

I thought the real life experience of surgeon Dr. Lim's experience during the time of beginning time of the Iraq war was astoundingly interesting and captivating. The fact that he had to leave at a moment's notice to be deployed, I just have a lot of a respect for him even if it seemed like he didn't have much of a say in the matter.

This information gave a good example of how anyone can become a "science person" if that's what they want. Dr. Lim specifically talks about his reputation in the science world, and how his instructors never would have believed he would get out there and do it.

This was a cool scientist spotlight because I'm very interested in the topic of obesity. I think its a serious problem that only getting worse as the article mentions, saying that weight loss surgery used to be something that was last resort and is not becoming the 1st choice to cure obesity. It's also something thats starting out younger and younger and helping lower or stop childhood obesity is something I'm very passionate about.

I've actually got a friend in the Special Forces currently and also another in the Air Force. So it feels good knowing they are in safe hands with doctors or surgeons like Robert.



Application Question 2

Pretend you are working in a team of health scientists to develop new heartburn medications. As we have seen, this is an important job, since persistent heartburn could signal damage leading to major problems, like esophageal cancer!

- One member of your team, Ashley, wants to create a medication that inhibits the activities of G Cells.
- Another member of your team, Babak, wants to create a medication that prevents chloride ion transporters from working. Discuss the merits of these two proposals.

From a cell physiology standpoint, how might Ashley's medication potentially prevent heartburn?

From a cell physiology standpoint, how might Babak's medication potentially prevent heartburn?

*"Ashley's medication will stop the production of a hormone called gastrin. Gastrin releases acid, so less gastrin means less acid. **Discuss the roles of parietal cells.***

*Chloride helps regulate pH, so with Babak's medication, there will be less chloride and less acid. **Discuss the roles of parietal cells. Discuss the role of HCl and how that is produced"***

How could we make this even more accurate and complete?

Application Question 1



Pretend a friend of yours, Llesenia, ate an apple this morning.

We then tried to figure out what happened to the atoms of Llesenia's apple and how that related to other organ systems you studied in Bio 40A/B.

For this question, Choose at least two (2) of the organ systems below and explain, from an A&P perspective, how they are involved in what happens to the atoms in Llesenia's apple.

Choose 2 to discuss:

- Nervous System
- Cardiovascular System
- Skin System
- Respiratory System

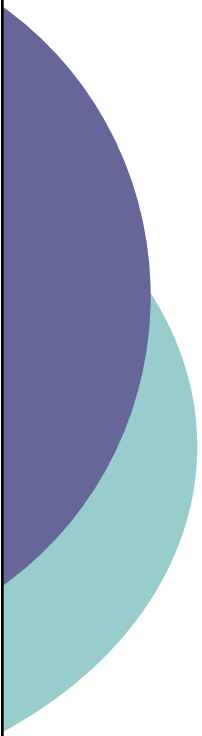
"Discuss things at the cellular or chemical levels.

Make connections to the process of digestion or absorption

Nutrients from the apple will help Llessenia's skin to stay hydrated and healthy.

Glucose from the apple will help the heart pump in the cardiovascular system."

How could we make this even more accurate and complete?



A Brief Preview of This Week

- Today

- A day extend our ideas about digestion to additional contexts

- Lab This Week

- Chemical Digestion

- Online Lecture Work This Week

- Finishing up digestion with the large intestine
 - Beginning to explore our next topic – the urinary system



What will you be able to do?

After class, you will be able to:

- Evaluate the physiological impacts and and potential complications surrounding various weight loss surgical procedures
- Predict how filtration in the kidney will change following mood-induced changes in blood pressure.

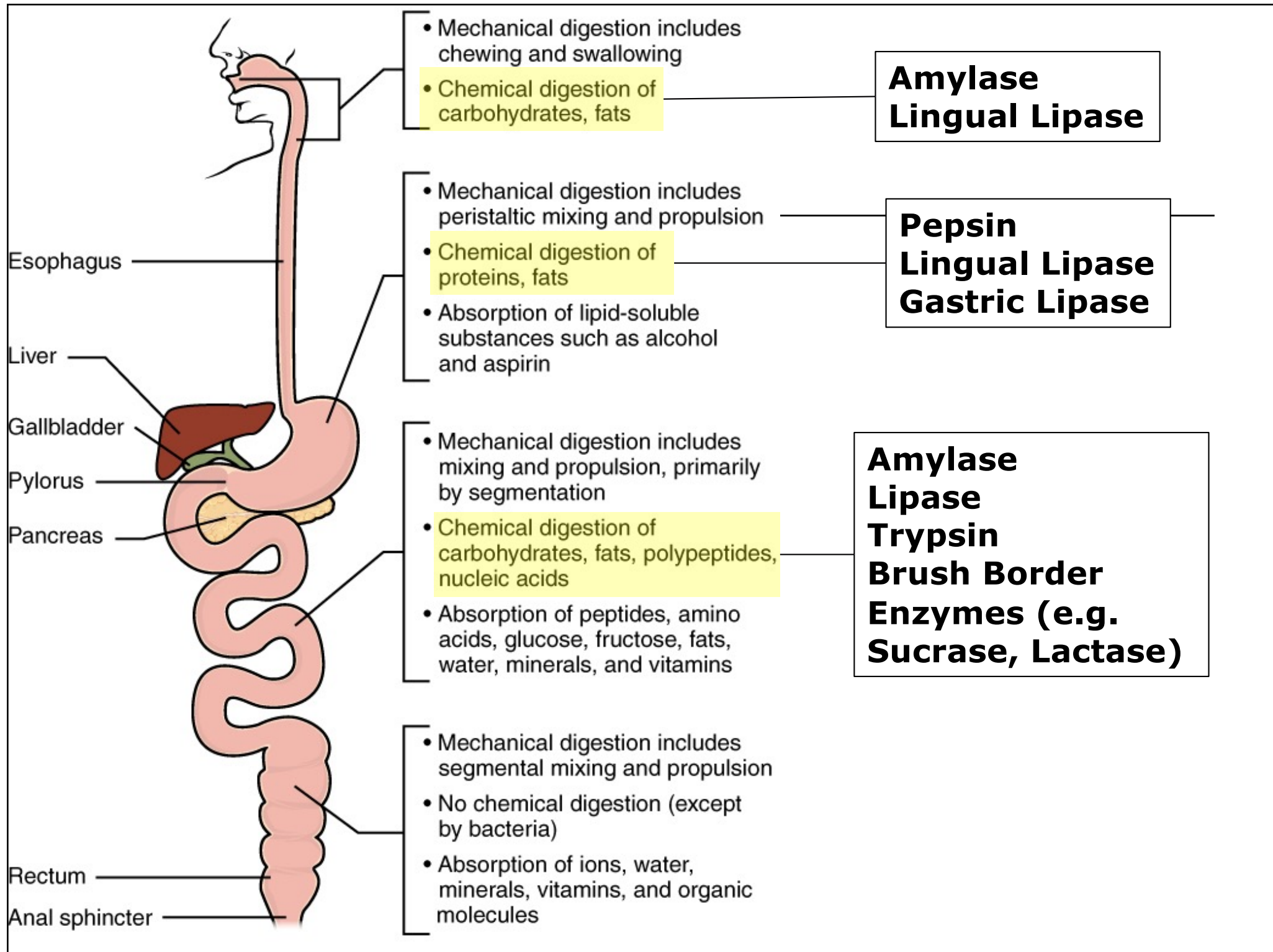
Considering your morning coffee...

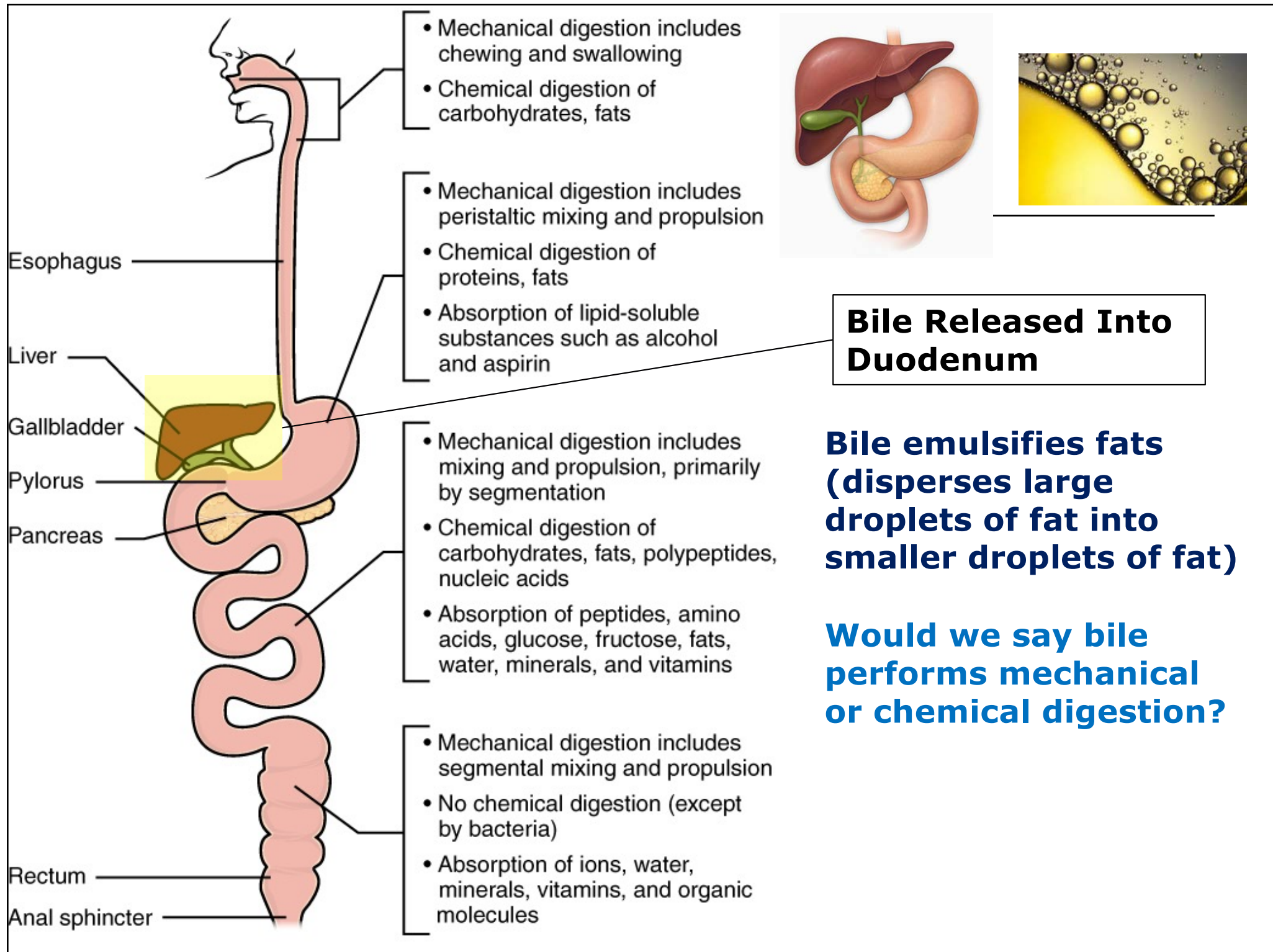


Pretend that you had coffee with cream and sugar this morning.

What happens to the following chemicals in this drink after entering your mouth?

- Carbohydrates from Cream/Sugar
- Fats from Cream
- Proteins from Milk
- Caffeine from Coffee
- Water

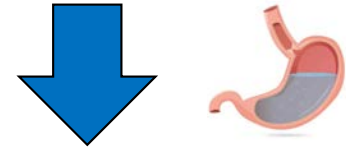




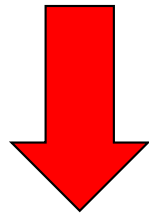
Water Molecules & Lipid Soluble Chemicals

Absorption

Absorption Begins In Stomach

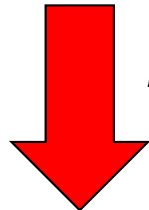


Simple Sugars



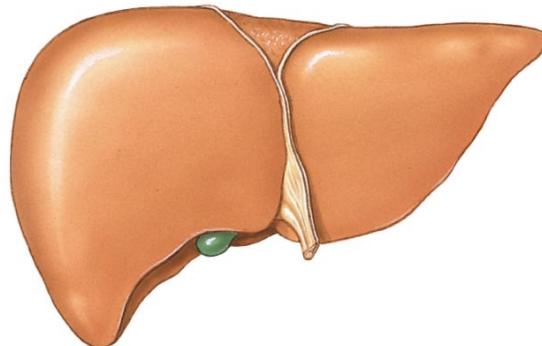
Capillaries in Villi

Blood

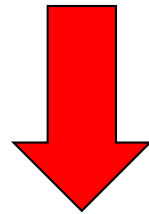


Hepatic Portal Vein

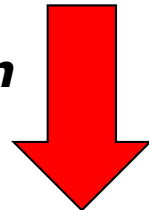
Directly to Capillaries in Liver



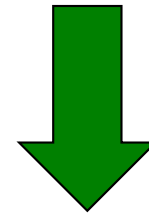
Amino Acids



Blood

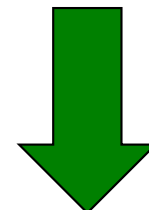


Fatty Acids



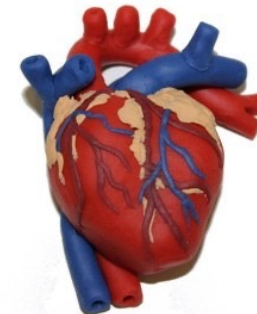
Lacteals in Villi

Lymph



Subclavian Vein

Right Atrium



Considering your morning coffee...



Pretend that you had coffee with cream and sugar this morning.

What happens to the following chemicals in this drink after entering your mouth?

How absorbed

- Carbohydrates from Cream/Sugar
 - Broken down by amylase (and other enzymes) and absorbed as simple sugars into capillaries in villi in small intestine
- Fats from Cream
 - Broken down by lipase fatty acids and absorbed in small intestine by lacteals into the lymph
- Caffeine from Coffee
 - Absorbed in stomach as caffeine



Considering the impacts of weight loss surgery

Benchmarking Best Practices in Weight Loss Surgery

You will now receive an excerpt from this academic article on one of four weight loss surgical procedures. Review the information and consider: **What, if any, impacts do you think that procedure would have on...**

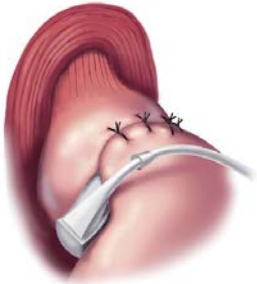


FIG 2. Adjustable gastric band. (Adapted with permission from Jones and co. Cine-Med Publishing, Inc., www.cine-med.com.) [Color version of figure is available online.]

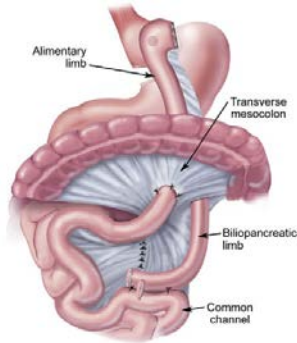


FIG 3. Roux-en-Y gastric bypass. (Adapted with permission from Jones and colleagues. Cine-Med Publishing, Inc., www.cine-med.com.) [Color version of figure is available online.]



FIG 8. Jejunoileal bypass. (Adapted with permission from Jones DB, Olbers T, Schneider B, Atlas of Metabolic and Bariatric Surgery, ©2010 Cine-Med Publishing, Inc., www.cine-med.com.) [Color version of figure is available online.]

Sleeve Gastrectomy

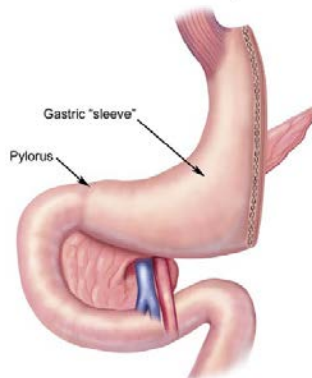
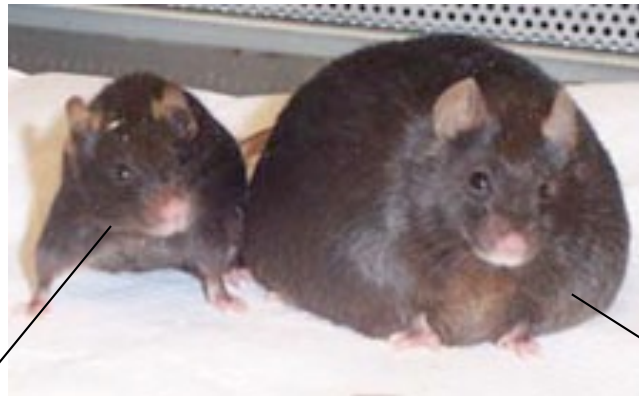
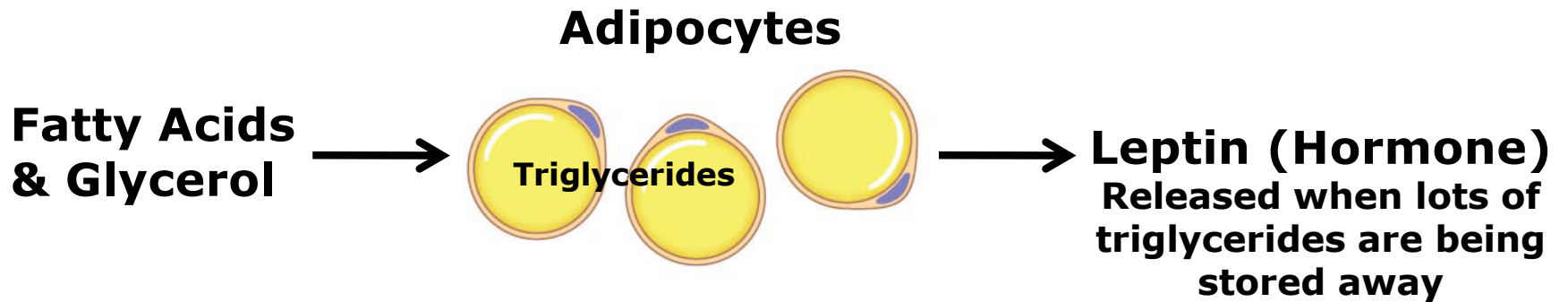


FIG 4. Sleeve gastrectomy. (Adapted with permission from Jones DB, Olbers T, Schneider B, Atlas of Metabolic and Bariatric Surgery, ©2010 Cine-Med Publishing, Inc., www.cine-med.com.) [Color version of figure is available online.]

1. Chemical Digestion
2. Mechanical Digestion
3. Absorption
4. Appetite/Nutrient Intake

Controlling Nutrient Intake



**Rat with Ability
to Make Leptin**

**Rat Unable to
Make Leptin**

What does this imply about how leptin affects nutrient intake?

Leptin Inhibits Appetite

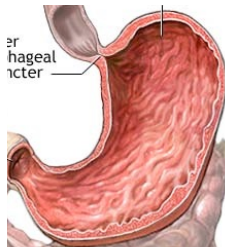
Nutrient Intake & Ghrelin (another hormone)

Ghrelin levels rise during fasting

Ghrelin levels drop if the stomach walls are stretched



What does this imply about how ghrelin affects nutrient intake?



Stomach Makes More Ghrelin When Empty



Stomach Makes Less Ghrelin When Full

Ghrelin Stimulates Appetite

Other factors impacting nutrient intake

Short-Term Regulation of Appetite

Stimulation of Satiety Center →

Elevated blood glucose levels depress appetite, and low blood glucose stimulates appetite. The likely mechanism is glucose entry stimulating the neurons of the satiety center.

Several hormones of the digestive tract, including CCK, suppress appetite during the absorptive state.

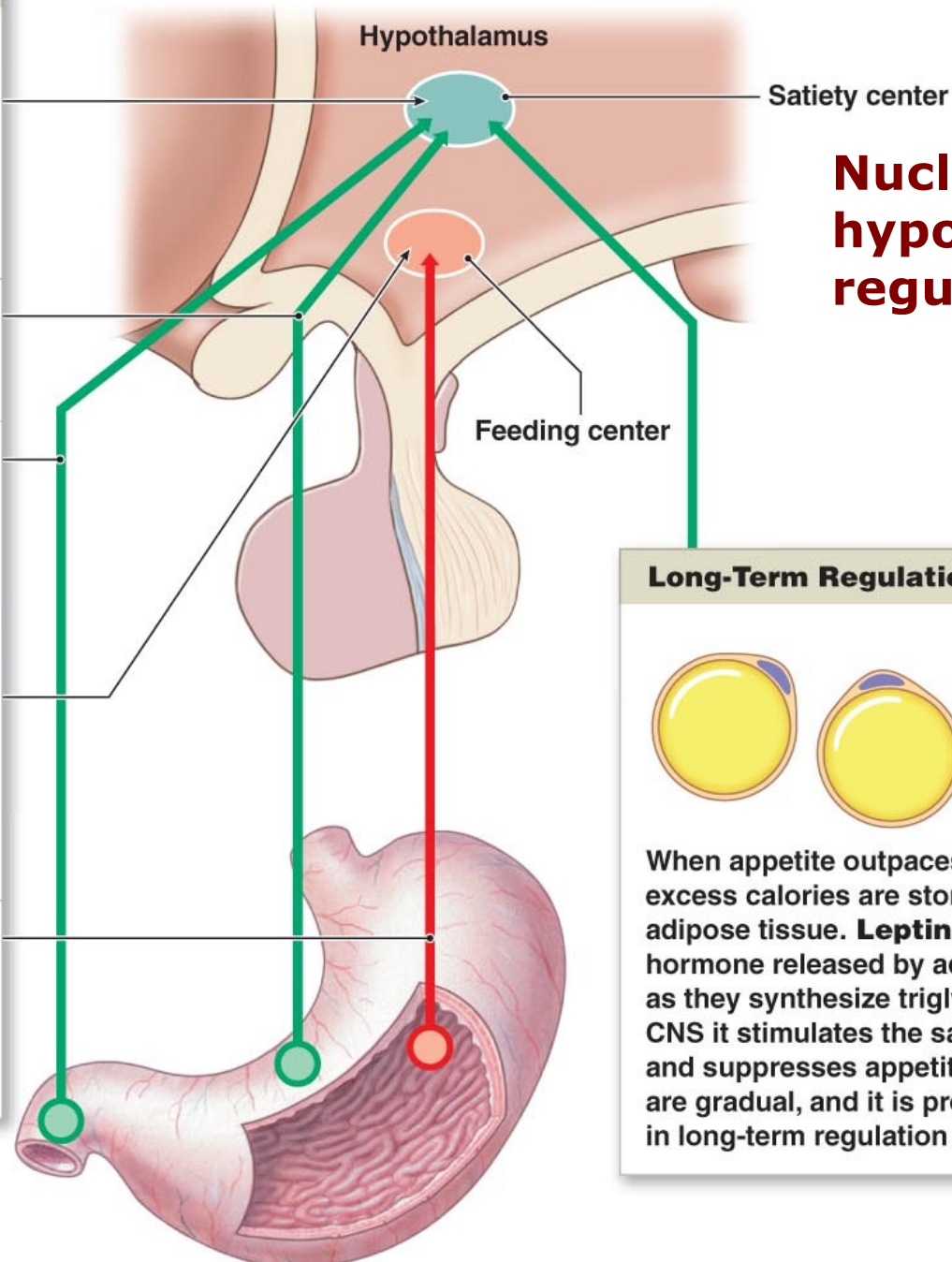
Stimulation of stretch receptors along the digestive tract, especially in the stomach, causes a sense of satiation and suppresses appetite.

Stimulation of Feeding Center →

Several neurotransmitters have been linked to appetite regulation. **Neuropeptide Y (NPY)**, for example, is a hypothalamic neurotransmitter that (among other effects) stimulates the feeding center and increases appetite.

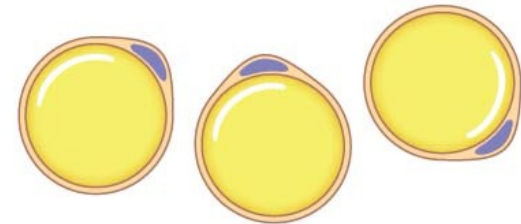
The hormone **ghrelin** (GREL-in), secreted by the gastric mucosa, stimulates appetite. Ghrelin levels are high when the stomach is empty, and decline as the stomach fills.

Mechanisms in the control of appetite



Nuclei in hypothalamus regulate hunger

Long-Term Regulation of Appetite



When appetite outpaces energy usage, excess calories are stored as fat in adipose tissue. **Leptin** is a peptide hormone released by adipose tissues as they synthesize triglycerides. In the CNS it stimulates the satiety center and suppresses appetite. The effects are gradual, and it is probably involved in long-term regulation of food intake.

Benchmarking Best Practices in Weight Loss Surgery

Robert K. Lim

Curr Probl Surg 2010;47:79-174.
0011-3840/2010/\$36.00 + 0
doi:10.1067/j.cpsurg.2009.11.003

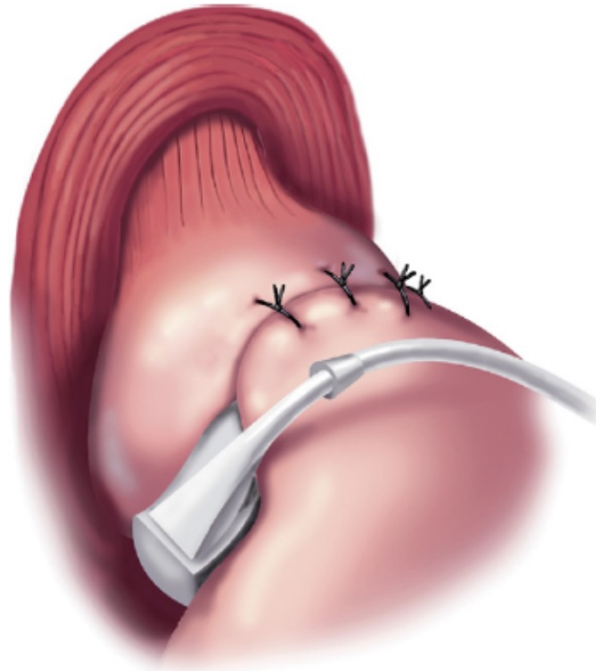


FIG 2. Adjustable gastric band. (Adapted with permission from Jones and colleagues,³³ ©2007 Cine-Med Publishing, Inc., www.cine-med.com.) (Color version of figure is available online.)

Adjustable gastric band (AGB) surgery is purely restrictive. It involves placing an inflatable Silastic band around the proximal aspect of the stomach to create a 30-mL gastric pouch³² (Fig 2). The band attaches via tubing to a port in the subcutaneous tissue, which can be accessed with a Huber needle, much like a chemotherapy port. The port is used by surgeons to inject or withdraw fluid from the band to further restrict or loosen it. Unlike for other restrictive procedures, research suggests that AGB does not result in changes in stomach hormone release.

- 1) Lipase and pepsin produced in stomach, so maybe putting this band will change the way the stomach makes chemicals like lipase and pepsin.
- 2) Creating this pouch in the stomach would affect mechanical digestion. Would impact mechanical digestion because it constricts the stomach.

3) Stomach absorbs water and some lipid soluble chemicals, so maybe the band will